

This is an excerpt from Frank Elavsky's dissertation on *Tool-making as an Intervention on the Accessibility of Interactive Data Experiences*, which can be accessed in full at this archival link:

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- Chapter 5: *Data Navigator*: Low-level Tooling for Creating Navigable Data Structures
- Chapter 6: *Skeleton*: Visual Authoring of Non-visual Data Experiences
- Chapter 7: *Cross-perception*: Rethinking Input Design Towards Blind Analytical Interaction
- Chapter 8: *Software*: Enabling Personalization of Interactive Data Representations for Users with Disabilities
- Chapter 12: The *Software* Option Taxonomy
- Chapter 11: Biographical Sketch

Abstract

In this dissertation, we contribute practical advancements in tool-making as an intervention on the accessibility of interactive data experiences. The thesis of this dissertation is as follows:

This dissertation argues that the tools practitioners use to build interactive data experiences are themselves sites where accessibility barriers are produced, prevented, or alleviated for both end users and authors. This work contributes five tools, Chartability, Data Navigator, Skeleton, Cross-perception, and Softeware, that collectively center accessibility work on the empowerment of disabled and non-disabled practitioners across the full arc of evaluation, data navigation, analytical interaction, and personalization.

Rather than framing accessibility research solely around ideal experiences for end users with disabilities, this thesis investigates why accessibility work is so difficult for the practitioners who build interactive data experiences and what tool-making can reveal about those difficulties. We organize this investigation around four domains where practitioners face the most persistent challenges: evaluation, navigation, interaction, and personalization. *Chartability*, a heuristic framework, contributed first and mapped the full landscape of accessibility barriers. *Chartability* helped us to identify and prioritize our three latter domains (navigation, interaction, and personalization) as the areas where the most severe gaps remain in practitioner work. *Data Navigator* and *Skeleton* then investigate navigation, finding that practitioners struggle because navigation structure has no visible, manipulable representation in their workflows. *Cross-perception* engages interaction, demonstrating that blind data analysis has been constrained by existing tools and that a new interaction design framework can reshape what analytical work is possible. *Softeware* addresses personalization, revealing that access needs genuinely conflict across users and that meaningful personalization requires system-level infrastructure that does not yet exist in practitioner tooling ecosystems.

Combined, these contributions provide empirical insights and practical advancements in the state of the art for tooling that bridges gaps in current accessibility practices in visualization and data science. Our work ultimately enables people with and without disabilities to better evaluate barriers in, analyze with, design for, develop, and personalize interactive data experiences. We demonstrate that tool-making is a productive intervention that both engages accessibility barriers and elucidates why those gaps exist in practitioner work.

Part I

Introduction

Chapter 3

Overview of Contributions

In this dissertation, we contribute practical advancements in tool-making as an intervention on the accessibility of interactive data experiences. The thesis of this dissertation is as follows:

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Rather than framing accessibility research solely around ideal experiences for end users with disabilities, this thesis investigates why accessibility work is so difficult for the practitioners who build interactive data experiences and what tool-making can reveal about those difficulties. We organize this investigation around four domains where practitioners face the most persistent challenges: evaluation, navigation, interaction, and personalization. *Chartability*, a heuristic framework, contributed first and mapped the full landscape of accessibility barriers. *Chartability* helped us to identify and prioritize our three latter domains (navigation, interaction, and personalization) as the areas where the most severe gaps remain in practitioner work. *Data Navigator* and *Skeleton* then investigate navigation, finding that practitioners struggle because navigation structure has no visible, manipulable representation in their workflows. *Cross-perception* engages interaction, demonstrating that blind data analysis has been constrained by existing tools and that a new interaction design framework can reshape what analytical work is possible. *Softerware* addresses personalization, revealing that access needs genuinely conflict across users and that meaningful personalization requires system-level infrastructure that does not yet exist in practitioner tooling ecosystems.

We engage each domain below with the questions: “Why does this work matter?”, “Why is it hard?”, and “What has tool-making within this domain showed us?”

The four domains of work that we engage in this thesis start first with **evaluation**. In work contexts where someone is designing and developing interactive data experiences, the practitioner must have the knowledge, tools, and resources available to systematically identify how their interfaces produce barriers for people with disabilities. A significant portion of professional accessibility work (arguably most, if not all) is founded on auditing and evaluating barriers to access. This work is pre-dominantly done through a standards-based approach [29], although in more robust evaluation work, people with disabilities are actively involved in the process [23].

Evaluation is difficult work because much of it is contextually defined by the author themselves, and most tasks at this intersection require careful, non-automated processes and meth-

ods [4, 23]. To make matters more difficult, no comprehensive guidelines, tests, and tools exist in any singular location. Practitioners often must gather these resources themselves, which tend to be situated towards accessibility in general or are high-level and provide minimal usefulness in practice. Additionally, practitioners themselves often have little knowledge about the veracity or quality of any given bit of information they gather [15, 27], and often do the work themselves to synthesize this disparate space of information into a usable format they can apply to their own evaluation work.

To engage this, our first main chapter focuses on *Chartability*, a heuristic framework that enables designers, developers, and auditors to systematically evaluate data visualizations and interfaces for a wide range of accessibility barriers, considering people with visual, motor, vestibular, neurological, and cognitive disabilities. In this project, we did the hard work for other practitioners and contributed our collection of synthesized accessibility resources in a single workbook. We had practitioners try out our resource in real environments, in-situ, in order to learn more about the challenges and barriers they themselves faced in evaluation work. With *Chartability*, practitioners, especially those with limited accessibility expertise, gained more confidence and clarity in assessing and improving their work. Additionally, *Chartability* has since become widely applied as a framework that isn't just used for evaluation but also as design guidance in many contexts, internationally, including policy organizations, governmental groups, and more than 100 companies and businesses.

Chartability then opened up a significant landscape of new projects and research directions. From a combination of my existing expertise as a visualization designer and engineer, in addition to our continued application of *Chartability* in the wild, we began to identify the trickiest and most-difficult domains of work for practitioners. *Chartability* has 50 total heuristics, or tests, each organized under one of seven principles. These three domains did not simply “emerge” on their own; they were evidenced by our applied work. Alongside the research activities that comprise this thesis, in a freelancing role we applied *Chartability* by partnering with industry, government, and non-profit organizations, conducting over 150,000 tests (both manual and automated) and working closely with dozens of designers and engineers. From that body of work, three larger domains stood out as the areas where the most severe and dramatic accessibility barriers remained unaddressed: data *navigation*, analytical *interaction*, and interface *personalization*.

So the next section of this thesis engages the first of these three: **navigation**. Navigation is a fundamental type of interaction that is leveraged by modern software-based assistive technologies. Screen readers, the primary tool used by people who are blind to interact with computers, navigate content. Additionally, many other assistive technologies, such as a sip and puff device (like the “POSSUM” from as far back as '63 [19]) also navigate. Navigational technologies are leveraged by people with a significant array of disabilities, yet tend to be entirely ignored by existing data visualization tools, which are pre-dominantly built to support direct input (using a computer mouse).

Empirical work has already demonstrated that structural navigation is actually good [34], even when regular alternative text (image descriptions) exist. This is both because people who are blind can gain both a high level understanding (from the description) as well as lower-level sense of the data's structure and arrangement, in addition to the fact that discrete, structural navigation exposes interactivity that may exist on any visualization elements (such as they can be hovered or clicked with a mouse in order to perform some action). So, if good empirical work

exists: *why haven't practitioners put this research into action? What makes this work hard to do?*

We first built *Data Navigator* to provide the building blocks we needed in order to address the technical and conceptual gaps that were required to make any visualization or visualization tool provide a navigable, interactive structure. *Data Navigator* is a low-level toolkit which can be used to construct accessible navigation structures such as lists, trees, and diagrams from an underlying graph structure. We leveraged graph theory for an applied HCI problem: nodes and edges represent any relationships within the data as a structure, which then supports rich expressiveness of data navigation experiences. Users can navigate discrete marks in a visualization, clusters, groupings, and more. In addition to its structural scaffolding, *Data Navigator* also supports a wide array of input modalities leveraged by people with disabilities (screen readers, keyboards, speech, and gestures).

Data Navigator provided a substrate, but this contribution alone wasn't enough to engage the question *why is navigation so hard, in practice?*. So the chapter following *Data Navigator* introduces *Skeleton*, a data navigation authoring tool built on top of *Data Navigator*. Our novel approach in *Skeleton* involves visualizing and making manipulable the nodes, edges, and textual data that comprise non-visual end user experiences. *Skeleton* visualizes the building blocks that comprise *Data Navigator*. Additionally, *Skeleton* provides expressive, rapid scaffolding capabilities that leverage data visualization rendering engines. This scaffolding engine helps practitioners quickly create common configurations for non-visual data navigation structures that retain visual congruence to the underlying structure.

But most importantly, *Skeleton* serves as a framework that shapes designerly consideration. Our conjecture was that because sighted practitioners cannot *see* navigation building blocks, they will not treat those elements as iterable design materials. We conjectured: Navigation is hard in practice because sighted designers face barriers to iteration and understanding. We conducted an empirical study with sighted practitioners and found that making non-visual elements visual helped practitioners shift from treating accessibility as a compliance task to treating it as a design problem, re-iterating on the visual aspects of their design, and engaging in the complex and nuanced components that comprise data navigation experiences.

Now, **interaction** becomes the next area we want to engage. Existing accessible data interaction for people who are blind, including our previous work on navigation (which is a form of interaction), predominantly seeks to expose information. This is what we call *access-oriented interaction*. In terms of low-level analytical tasks, most are then made feasible through navigation, sonification, or summarization-based and question-answering approaches: retrieving values, filtering, computing derived values, sorting, determining ranges, clustering, and finding outliers. What remains are analytical tasks that, despite being “low level” (understood as *unable to be reduced into more fundamental tasks*), are cognitively highly complex: finding correlations and characterizing distributions [1]. These tasks require complex hypothesization and exploration, rather than a system that simply encourages surfacing what is known or what is already present in the data: it requires combining, remixing, restructuring, and dividing data.

But blind *analytical interaction* isn't just important to engage because it is understudied, it is important to engage because many of interactive information visualization's most impactful tools for data science enable it [5, 9, 22, 30]. In visualization, *cross-filtering* is one example of an interaction that enables a user to filter one visual space while seeing a coordinated change in

another visual space simultaneously and near-instantaneously. The speed of input interaction and perception of output also matters: even a small bit of latency changes the quality of a user’s data exploration activities [17]. We conjectured that a screen reader, the most-used tool leveraged by blind people when interacting with computers, may be insufficient for engaging this task.

To engage this, this thesis introduces *Cross-perception*, an approach for building analytical interactions that support perception in one space of input interaction with simultaneous, non-competing perception of output in another space of data representation. We first formalized a design framework for producing *cross-perception* experiences and then built a novel prototype device, the *cross-feelter*, that enables blind *cross-perception* of a cross-filtering data exploration interface. In an empirical study with blind users (with and without existing data expertise), we found *cross-perception* speeds up analytical exploration by 90% and helps blind users consider vastly more questions of their dataset (+188% computational queries, +54% spoken aloud) compared to a screen reader-driven interaction.

Beyond performance, we found that our input modality itself shaped the character of analytical engagement: participants didn’t just work faster, they considered more dimensions of their data and asked qualitatively different questions. The *cross-feelter* also reduced anxiety and substantially increased enjoyment, particularly for participants without prior data expertise, suggesting that the barriers blind practitioners face in data work are not only functional but affective. Additionally, we had our blind practitioners imagine new interaction possibilities that *cross-perception* could enable including and beyond our *cross-feelter* device.

Our final domain of work is the most difficult for visualization practitioners to engage: **personalization**. While *navigation* demands better software tooling and visual support and *interaction* requires new hardware, *personalization* completely re-orientes how software authoring takes place. Personalization matters because of *access friction*, which is a design challenge where one design or interface configuration that might be accessible for one person or group of people turns out to create barriers for someone else [7, 11]. In existing work on personalization and accessibility, studies have demonstrated that end-user control is great to have and can alleviate friction [14, 33], but little work has been done to explore what personalization looks like for an existing data visualization library and how practitioners should build and maintain their existing systems to support it.

Our final chapter introduces *Software* to address the tension between standardized accessible design and the diverse needs of real users with disabilities. In the wild, *access friction* exists in every design that reaches a public audience; it is inevitable. This tension ultimately means that some users have a worse experience, and may even face exclusion, with any particular design configuration. So for this work, we conducted our research in-situ with visualization software engineers and designers and worked to build a scalable, flexible software system dubbed “softerware” that enables end users to manipulate the appearance and functionality of the charts and graphs they encounter according to their own preferences. We conducted empirical research to inform our collaborators, as well as other visualization system authors, with guidelines and considerations for building *softerware* systems. In our study, no two participants chose the same preference configuration and participants with the same diagnosed condition sometimes needed opposite design treatments. Practitioners, meanwhile, immediately raised ethical concerns about whether personalization would let designers off the hook for poor defaults. These findings revealed that the real barriers to personalization are not at the level of any individual chart but at

the level of system infrastructure: without persistence, cross-system interoperability, and shared standards, the effort required of end users exceeds the value they receive.

Combined, these contributions provide empirical insights and practical advancements in the state of the art for tooling that bridges gaps in current accessibility practices in visualization and data science. Our work ultimately enables people with and without disabilities to better evaluate barriers in, analyze with, design for, develop, and personalize interactive data experiences. We demonstrate that tool-making is a productive intervention that both engages accessibility barriers and elucidates why those gaps exist in practitioner work.

Part VI

Conclusion

Chapter 9

Findings

The five projects in this thesis were presented as five chapters, each with its own results and its own conclusion. Read separately, they contribute a heuristic resource for evaluation, a structural toolkit for navigation, an authoring environment for non-visual design, an interaction technique and device for blind analysis, and a system and study of end-user personalization. This chapter states what they show *together*. Chapter 2 closed on a lopsided field: a breadth of empirical work and exemplary prototypes on one side, and a near absence of practitioner tools in production environments and practitioner action on the other, with the hardest remaining challenges concentrated in navigation, interaction, and personalization. The findings below are what this thesis learned by working inside that gap.

Each finding is a synthesis across chapters, and each is traceable to results reported in the body of this thesis. We have written them for two audiences at once: practitioners who build, evaluate, and maintain data experiences, and researchers who study how that work happens. Where a promising idea outran the evidence, it is not here; it is in the next chapter, with the rest of the discussion and future work.

9.7 What Tool-making Did Not Do

The findings above would be misread without their boundaries. Three are most apparent in this thesis, which we unpack below.

9.7.1 State of the Art does not Necessitate Adoption

First, a tool’s existence, even if it is the “best” system by some argument or measure, did not cause its adoption. This should seem obvious, and yet technical research appears to operate with an unspoken assumption that *something* practical should happen at scale after a clever or novel prototype is produced. Research ideas are brought into the world and largely abandoned almost immediately [12]. *Data Navigator* was (and perhaps still is) the most flexible and expressive system made for data navigation, and yet translating designs into it remained too complex to simply hand off, which is what motivated our industry collaborations and ultimately *Skeleton*; *Software*’s participants made clear that personalization that must be redone per chart is not worth their effort; and the proportion of the web’s most popular pages with critical accessibility failures is essentially unchanged across the six years spanned by this research [31]. Chapter 2 observed that a practitioner cannot simply install a paper; this thesis adds that practitioners will not, and often cannot, adopt even a working tool absent authoring support, incentives, infrastructure, and policy around it. Existence is the beginning of an argument for adoption, not the conclusion or even motivation for one.

9.7.2 Tooling does not *Solve* Access Friction

Second, tools did not solve access friction, and our evidence suggests they may not be able to. Access friction may simply be inevitable, as others already demonstrate and argue [8, 11, 20]. In the *Softerware* study, no two participants chose the same configuration, participants with the same diagnosed condition needed opposite design treatments, and one participant wanted strong defaults rather than any customization at all. Personalization *relocates* friction and gives people agency within it; but it does not engineer the friction away. Strong accessible defaults, despite being a source of friction in the first place, remain a prerequisite of personalization.

9.7.3 Authorship by People with Disabilities Remains Unresolved

Third, the tools in this thesis did not redistribute authorship. They mostly serve practitioners building *for* disabled users: *Skeleton*'s study evaluated sighted practitioners' engagement, not the experiences their structures produce for blind users, and while *Cross-perception* and *Softerware* begin to shift who holds the interaction, people with disabilities remain largely excluded as authors, analysts, and people with power over data interfaces. This marks what should be expected from tooling interventions and what should be expected from policy, practice, and the sharing of power [28]. Some existing work has used tooling as a space to explore what de-centering visual authoring might look like [35], but this largely remains an area with significant gaps in practice. The next chapter takes up exactly these questions, of what tools are downstream from, who is responsible for repair, and what this work asks of the field beyond its tools. We view these gaps as generative spaces for future work.

Chapter 10

Discussion & Future Work

10.3 Who is responsible for repair?

Lastly, we want to revisit one of this thesis’s opening points, where we argue that the *tool-makers* are first responsible for repair. This is true. However, the most pressing issue we have faced in recent years is mostly unmentioned across these research projects: tool-makers might be responsible, but this is because they are the only ones who have the *power* to make things accessible. Does this always need to be the case? Can we imagine an artifact’s authority over the user’s ability to access being designed towards self-subversion [6] or de-centralized agency [3, 16, 21], instead? What might that look like, concretely?

In *Software*, we begin to engage this larger problem in terms of an idealized state where a user can repair or re-design their own experiences. But to me, this self-repair is like laying down train tracks for yourself as you move a locomotive, but then lifting up your own tracks behind you as you go. You’re the only one helping yourself. This is not ideal, for you or others.

What we need are broad, lasting, infrastructural changes. On the web, this problem becomes quite difficult to solve. A personal computer or device? Again, someone can auto-design their interfaces into a better state. But back when we started *Chartability*, the WebAIM Million’s report showed more than 95% of the top one million website home pages contain at least one critical accessibility error. And now, more than six years later, that proportion is unchanged [31].

Some had imagined that generative AI would solve the massive infrastructural repair problems we face. But unfortunately, the latest WebAIM Million report shows that since 2020, ARIA usage has increased and correlates to more errors, while use of `tabindex` on a page has increased nearly 300% and also correlates to more errors on a page. If anything, during the age of generative AI, we have seen existing bad patterns worsen in prevalence and complexity.

We firmly believe that a tools-based approach is not enough on its own. Tool-making cannot be the *only* intervention on inaccessibility. Tools and tool-making, as our thesis argues, have a powerful role to play. But we simply can’t tool our way out of failed infrastructure and inadequate policy when someone else *owns* the tools and tool-making. Visiting a website is like going into someone else’s home: arranged according to their effort, tastes, and so on. If you can’t access their home, you essentially need to request that they let you in personally. Website repair always falls to the owner and maintainer of a website, and they largely don’t take any meaningful action.

Sidewalks outside of homes are a good parallel to this problem. Sidewalk accessibility is a massive infrastructural problem [25], and yet localities treat sidewalk maintenance in different ways: some, like where I, Frank, presently live in the south hills of Pittsburgh, put the onus on the homeowner whose house and property the sidewalk touches. In other places, sidewalks are considered a public path, similar to a roadway, and are maintained through public tax and resource management. To no surprise, privately-maintained, public-access sidewalks are worse for people in pretty much every way than publicly-maintained ones [32]. This is because private homeowners don’t care about sidewalk maintenance unless the city manages to fine them or they get sued.

And the web is a collection of private spaces that you visit privately. There is no truly shared, universally democratic, public space on the web. Centralization is partly to blame: sharing space while scaling leads to consolidation.

So our future work will continue to wrestle with the same tensions of scale, repair, and anti-consolidation of power, motivated by the same WebAIM Million report. But now we look to questions of *democratic* and *radical* access to accessibility repair. The barriers we hope to tackle in the future are political and infrastructural. Perhaps tool-making will participate in this work, but it seems clear now from our work that the upstream technical problems and socio-political conditions that tools inherit, will likely not be addressed by tools alone.

What does “democratic” and “radical” infrastructure work look like? It will probably be an extension of *Softerware*, to some degree. We imagine future research that explores public-first spaces, ones where access is socially negotiated and repair belongs to all of us. Is this an autonomous space, like an autonomous zone [2] separate from the web? Above it, looking down into it, like shared annotation tools but capable of sharing the manipulation of websites [24]? A space with ambient co-repair, modeled after projects that bring people together [26] or that allow community “fixing” of misinformation [13]? Perhaps feminist thought on the ethics of care can help us [10, 18]? Or maybe it will be something else entirely; I’m not yet sure. But what made the web fantastic years ago is long gone; most of it has been hedged into corporate spaces that are controlled, maintained, and repaired by corporate power. And these entities are notoriously bad at repair. What we imagine in the future involves reclaiming a sense that the web is *ours*, belongs to *us*, and that ultimately *we* are responsible for making it accessible.

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